

Skills Summary

Place Value with Base-Ten Models

Use this reference while completing your worksheet or when studying.

Skill 1 — Reading a base-ten model

Block values:

- a **flat** (big square) = 1 hundred
- a **rod** (tall bar) = 1 ten
- a **cube** (small square) = 1 one

Count each kind of block, then add the values.

Example: A model shows 2 flats, 3 rods, and 4 cubes. What number is it?

Step 1. Count each size of block separately: flats are hundreds, rods are tens, cubes are ones.

Step 2. $2 \text{ hundreds} + 3 \text{ tens} + 4 \text{ ones} = 100 \cdot 2 + 10 \cdot 3 + 4 \Rightarrow$ **234**

Try it: A model shows 3 flats, 1 rod, and 6 cubes. What number is it?

check: 316

(!) Watch out: If a model has *no* rods, the tens place still needs a digit — write a zero there. Three flats and eight cubes make *three hundred eight*, never thirty-eight.

Skill 2 — Hundreds, tens, and ones in a number

Rule: Each digit's place tells its value. In a three-digit number, the first digit counts **hundreds**, the middle digit counts **tens**, and the last digit counts **ones**.

Example: How many hundreds, tens, and ones are in 358?

Step 1. Read the digits from left to right and name each place.

Step 2. The 358 splits by place: hundreds digit, tens digit, ones digit. $\Rightarrow$ **3** hundreds, **5** tens, **8** ones

Try it: How many hundreds, tens, and ones are in 726?

check: 7 hundreds, 2 tens, 6 ones

Skill 3 — Comparing two models

Rule: Compare place by place, biggest first.

- First compare **hundreds**. More hundreds wins.
- If they tie, compare **tens**. If those tie too, compare **ones**.

Example: Which is greater, 341 or 314?

Step 1. Compare hundreds first: both numbers have three hundreds, so move to the tens.

Step 2. 341 has four tens and 314 has only one ten, so 341 is greater. $\Rightarrow$ **341 > 314**

Try it: Write $<$, $>$, or $=$: compare 562 and 526.

526 < 562 ✓